Sim-to-Real Transferability and Calibration Requirements for Connected-Vehicle Road Roughness Estimation

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***Abstract*—Connected cars could monitor road roughness continuously, but mapping their vibration to the International Roughness Index (IRI) requires paired drives over profiled roads, and vehicle simulation over existing profiles can generate such pairs. We quantify what transfers from simulation to a production car on public profilometer-referenced data (LiRA-CD, 168 km), testing on held-out routes with spatial-block bootstrap intervals. A randomised half-car simulator driven over measured profiles trains gradient boosting that ranks road segments within routes about as well as real training (Spearman 0.82–0.84 vs 0.83 under nine sensor and vehicle models), yet an untrained, speed-normalised vibration RMS ranks within 0.05 of both on every route: ranking is not the bottleneck. The bottleneck is the absolute level: the zero-shot bias ranges from -0.09 to $+0.43$ m/km and depends on how well the simulated sensor path reproduces the car's vibration above 10 Hz and on the simulated vehicle population. Five kilometres of paired drives resolve it. Simulation with a level offset (pooled MAE 0.42–0.56 m/km) or joint training (0.45–0.47 m/km) beats training on those drives alone (0.68 m/km) for every variant, even with one pass per segment, and the reference simulator with a 5 km offset (0.42 m/km) outperforms 20 km of real drives alone (0.46 m/km). Game-engine data, whose ray-cast road elevation carries 42–54 times the short-wavelength power of measured urban roads, do not transfer.**

***Index Terms*—International Roughness Index, connected vehicles, vehicle vibration, sim-to-real transfer, calibration, road profile.**

I. Introduction

The International Roughness Index (IRI) is the standard summary of longitudinal road roughness [1], [2]. It is computed from a longitudinal profile measured by a laser profilometer, an instrument too costly to survey minor roads often. Production cars that already carry accelerometers could monitor roughness continuously [3], [4], but their vibration depends on the vehicle, the sensor mounting and the speed. The mapping from vibration to IRI must therefore be established for each car type from paired data, drives of that car over road sections whose profile has been measured [5], and collecting such drives is the expensive step.

Vehicle simulation offers another source of pairs: driving a vehicle model over profiles that road agencies already survey yields vibration with exact IRI labels, at any speed and for many vehicle parameters. Convolutional networks trained on simulated responses of four vehicle types estimate IRI in simulation [6]; regressors trained on simulated responses were tested with a probe vehicle for the weighted longitudinal profile [7], as described in [8]; networks trained on a full-vehicle model over slightly modified measured profiles were applied to that vehicle's measured responses [9]; and simulation-trained classifiers of roughness grades were tested on a real vehicle [10]. Calibrated vehicle models instead invert car vibration into road profiles [11], [12] or derive an IRI–vibration relation by simulation [13], and learned estimators trained on real drives [14], [15], including on the data used here [8], need paired data for every car type. In robotics, randomised simulation [16], [17] adapted with a few real roll-outs [18] is the standard route from simulation to reality. What has not been shown, for continuous IRI from a production car on held-out roads, is which parts of a simulation-trained estimator transfer, which simulator choices set its accuracy, and how much paired real data must accompany the simulation.

We answer these questions on LiRA-CD [19], public data of one car model on four routes with profilometer references. Estimators trained on real drives are compared with the same estimators trained only on simulated drives over the measured profiles of the other routes, under nine sensor and vehicle models of the simulator, and with the simulation combined with k km of paired drives (Fig. 1). Our contributions are:

- We show that ranking segments within a route is not the bottleneck: an untrained, speed-normalised vibration RMS ranks them within 0.05 in Spearman correlation of the best trained estimators, real or simulated (Section III-A).
- We show that the absolute level is the bottleneck, and that it depends on how well the simulated sensor path reproduces the car's vibration above 10 Hz and on the simulated vehicle population (Section III-B).
- We quantify the calibration requirement: about 5 km of paired drives, with one pass per segment, fix the level, and joint training on simulated and real drives stays better than real-only training up to 20 km for every simulator

variant (Section III-C).
- We show that simulation must be driven by measured profiles: game-engine road meshes yield an implausible IRI ground truth and do not transfer (Section IV).

Code, configurations and predictions will be released with the published version.

II. Data, Simulation and Estimators

Real drives and labels. LiRA-CD [19], [20] contains Renault Zoe cars instrumented with an AutoPi telematics unit (three-axis accelerometer, GPS, CAN bus) fixed to the frame in the middle console near the front axle [19], and P79 laser profiles of the same lanes. We use the four routes with car data, two urban roads in Copenhagen (CPH1, CPH6) and two motorways (M3, M13), in both directions; the accelerometer is logged at 52.6 or 333 Hz and speed comes from the CAN bus. After map matching to the P79 chainage, the data contain 16,293 segment-passes over 1,678 segments of 100 m (168 km), about ten passes per segment. Labels are the IRI per 100 m (mean of both wheel paths), recomputed from the P79 profiles with an implementation of the quarter-car algorithm that reproduces the profilometer's own IRI ($r = 1.000$). Urban segments average 3.36 m/km at 43 km/h, motorway segments 1.16 m/km at 87 km/h; the surveys preceded the drives by two to fifteen months.

Simulator. A linear four-degree-of-freedom half-car (bounce, pitch, two unsprung masses) is driven by the mean of the two wheel-path profiles under the front axle and the same profile delayed by the wheelbase under the rear axle, after a 250 mm tyre-envelope average; simulated units are labelled like the real ones. Vehicle parameters are drawn from passenger-car ranges [21] (sprung mass 1000–1800 kg, body frequency 1.0–1.6 Hz, damping ratio 0.20–0.45) and speed independently of roughness (20–110 km/h). A sensor model adds an optional holder resonance, anti-aliasing, a mounting rotation, irregular sampling, scale and bias errors, noise, quantisation and sample loss. The reference simulator **V0**, fixed before any test evaluation, uses 30 random vehicles over the measured profiles of the training routes and a 50/50 mixture of a telematics-like sensor (55–62 Hz, 0.01 g, rigid) and a phone-like sensor (90–210 Hz) with a holder resonance at 15–40 Hz. Eight variants change one ingredient: either half of V0's sensor mixture alone (**V1** telematics-like, **V2** phone-like), the missing cells of a sensor-family $\times$ holder design (**V7**, **V8**), an ideal sensor (**V3**), a telematics sensor matched to statistics of the LiRA-CD streams alone (**V5**) or mixed with the phone-like sensor (**V6**), and one mid-range vehicle instead of 30 (**V4**); **V9** replaces the measured by synthetic ISO 8608 profiles [22]. Simulated drives never use the profile of the test route.

Features and estimators. Real and simulated accelerometer streams pass the same code: resampling to 200 Hz, zero-phase 20 Hz low-pass, decimation to 50 Hz, projection on the pass-mean gravity direction (orientation invariance) and a 0.3 Hz high-pass. A unit is one pass over one 100 m segment, described by its mean speed and speed variation and by 11 features of the dynamic vertical acceleration: log RMS, peak statistics, crest factor, kurtosis, RMS in the 0.5–2, 2–5, 5–10 and 10–15 Hz bands, above 1 Hz and at wavelengths below 5 m. The estimators predict log IRI: a road-class prior (map only); LightGBM [23] on speed only; a vibration index $\log \mathrm{IRI} = a + b \log(\mathrm{RMS}/v^{\alpha})$; a calibrated quarter-car inversion, which fits a quarter car to the car's empirical transmissibility on the training routes and inverts each unit's spectrum through the Golden Car (it reproduces the IRI of true profile spectra with $r = 0.94$); LightGBM on all features; an MLP; and a 1D-CNN on the resampled acceleration (five-seed ensembles). The untrained statistic $\mathrm{RMS}/v^{0.5}$ serves as a ranking baseline. Simulation-trained (zero-shot) estimators are fitted to the simulated units of one variant. LightGBM trained on whole real training routes selects one of six configurations by inner leave-one-route-out validation; all other LightGBM models use the first configuration.

Paired data. Paired data are k km of training-route road, drawn as contiguous 1 km blocks (20 draws per k, shared by all methods), with all real passes over them: at $k = 5$, 475 segment-passes on average, about 47 km of driving over 4.9 km of road. To test whether this density is needed, each segment also keeps at most $m \in \{1, 2, 5\}$ passes. The paired passes are combined with LightGBM trained on simulated drives over all training-route profiles by a level offset (median log ratio of label to prediction), an affine map in log space, or joint training on simulated and real units with equal total weight; the comparators are LightGBM and the vibration index trained on the paired passes alone.

Evaluation. Each route is held out in turn, both directions together, and nothing from it enters training, calibration or model selection. A segment is predicted by the mean of its pass predictions. We report MAE, bias and Spearman's ρ; *within-route* values average the four routes, *pooled* values cover all 1,678 segments, and the *level* of a road class is the mean log ratio of prediction to label over its segments, averaged over its two routes. Segment errors are spatially correlated (autocorrelation 0.44 at 100 m), so paired differences are bootstrapped over 5 km blocks (40 clusters; single-estimator spectra and elasticities: 1 km blocks). The estimators, the simulator V0 and the evaluation were fixed before any test result; the variant, calibration and spectral analyses were designed after the test routes had been evaluated and are exploratory.

III. Sim-to-Real Transfer and the 5-km Calibration Requirement

A. Ranking transfers, the level does not

Table I compares the estimators on the held-out routes (per-route values and paired intervals: Supplement S4). Real-trained LightGBM reaches a within-route MAE of 0.39 m/km; its MAE per road class is 19 % (urban) and 12 % (motorway) of the class's mean IRI. The quarter-car inversion ranks segments as well ($\rho = 0.82$) but errs more (paired difference $+0.074$ [$+0.039$, $+0.105$] m/km), and the 1D-CNN is worse by $+0.045$ [$+0.019$, $+0.075$] m/km, almost entirely on one urban route. Trained only on simulated drives, LightGBM

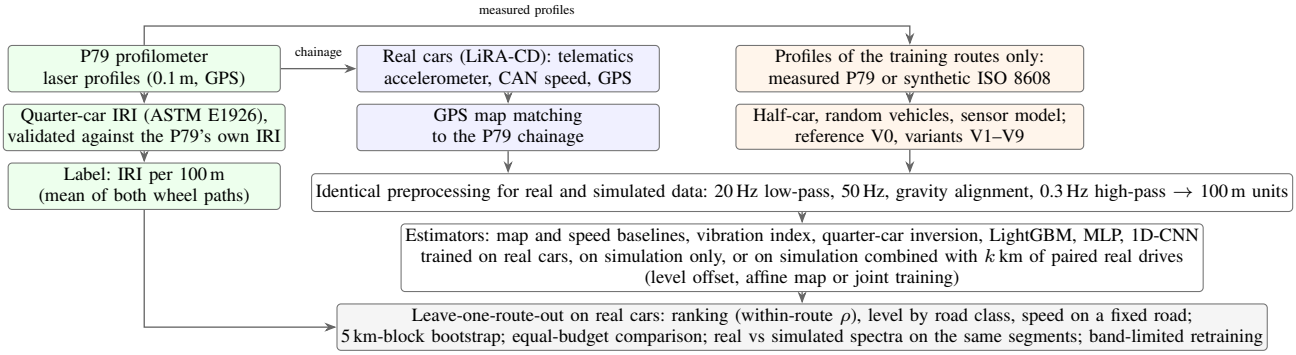


Fig. 1. Study design. Real and simulated units are labelled by the same IRI code, validated against the profilometer's own output, and preprocessed identically; every estimator is evaluated on real routes it never saw, and simulated drives never use the profile of the test route.

TABLE I
HELD-OUT ROUTE PERFORMANCE ON LIRA-CD (100 M SEGMENTS). MAE AND SPEARMAN ρ WITHIN ROUTES (MEAN OF THE FOUR HELD-OUT ROUTES), BIAS POOLED OVER ALL SEGMENTS, IN M/KM. SIMULATION-TRAINED MODELS USE NO REAL LABELS; GAME ENGINE: LABELS FROM THE RAY-CAST MESH ELEVATION AFTER SMOOTHING (SECTION IV).

Estimator	MAE	ρ	Bias
Without vibration or without training			
Road-class prior (map)	0.75	–	−0.20
LightGBM, speed only	0.69	0.39	−0.33
RMS$/v^{0.5}$, untrained	–	0.82	–
Trained on real drives of the three other routes			
Vibration index	0.62	0.76	−0.13
Quarter-car inversion	0.46	0.82	−0.08
1D-CNN	0.43	0.81	−0.21
LightGBM	0.39	0.83	−0.11
LightGBM trained on simulation only			
Measured profiles, V0	0.42	0.83	+0.01
Measured profiles, V0–V8	0.38–0.67	0.82–0.84	−0.09 to +0.43
Synthetic profiles, V9	0.71	0.81	+0.36
Game engine	1.26	0.35	−1.27

with the reference simulator V0 reaches 0.42 m/km, a paired difference to real training of +0.019 [−0.003, +0.040] m/km, and ranks segments alike ($\rho = 0.83$ vs 0.83).

Ranking, however, is not the hard part. A Spearman correlation within a route is blind to any monotone calibration, and the untrained statistic RMS$/v^{0.5}$ reaches $\rho = 0.82$: on every route it is within 0.05 of real-trained LightGBM, the MLP, the CNN, the inversion and simulation-trained LightGBM under every variant (real-trained LightGBM ranks the urban routes better by +0.042 and +0.044 and the motorways worse by −0.020 and −0.023), whereas the vibration index, LightGBM without speed inputs and the speed-only model rank up to 0.10, 0.18 and 0.72 worse (Table S4).

What must be learned is the level. Across the nine measured-profile variants V0–V8 the within-route ρ stays at 0.82–0.84, while the pooled zero-shot bias ranges from −0.09 to +0.43 m/km (Fig. S4). Even V0 matches the level only on average, combining a matched urban level (log level +0.006) with a motorway over-prediction of 16 %. In the sensor-family × holder design, adding the holder resonance shifts the bias by −0.35 and −0.36 m/km and switching the sensor family by −0.16 and −0.17 m/km, while replacing V0's 30 vehicles by one mid-range vehicle (V4) with the same sensors raises it by +0.39 m/km: the level depends on both the sensor path and the vehicle population.

B. Where the simulated vibration differs

On the same segments and speed bands, the real car vibrates more than every simulated variant at 2–5 Hz (2.4–3.6 times the power) and like them at 5–10 Hz, but above 10 Hz the variants differ (Fig. 2a): at 10–15 Hz the real car has 5.2 and 4.9 times the power of the ideal and the rigid telematics-like sensor, yet only 1.9 times that of the phone-like sensor with holder resonance. Across the nine variants the zero-shot bias follows the amplitude deficit at 10–15 Hz most closely (Spearman 0.97; Fig. 2b), but also that at 5–10 Hz (0.80) and of the overall RMS (0.88), and hardly those at 0.5–2 and 2–5 Hz (0.22, 0.25). V4 departs from this pattern: its 10–15 Hz ratio is close to V0's (1.93 vs 1.86), its bias is not (+0.40 vs +0.01 m/km).

To test which inputs carry these differences, we retrained every variant on the five features without vibration above 10 Hz. Over the eight sensor variants, the spread of the mean log level fell from 0.25 to 0.08 (32 %; Fig. 2c), whereas dropping only the 10–15 Hz band changed it by at most 2 %: the sensor variants differ through all features that carry vibration above 10 Hz. The low-band estimator is less accurate (pooled MAE after a 5 km level offset 0.48–0.57 instead of 0.42–0.56 m/km), and V4 stays outside the range of the sensor variants, so the vehicle population sets part of the level independently. This test acts on the estimator's inputs; the physical source of the car's extra vibration above 10 Hz, where its empirical transmissibility peaks (12.6 Hz), is not identified, and the linear half-car omits candidates such as local modes of the console mount, tyre and suspension nonlinearity, drivetrain vibration and roll. The simulated vibration also grows faster with speed than the real car's on the same roads (elasticity 0.92–1.30 vs 0.60 on urban and 1.07–1.40 vs 0.85 on motorway segments), for every variant.

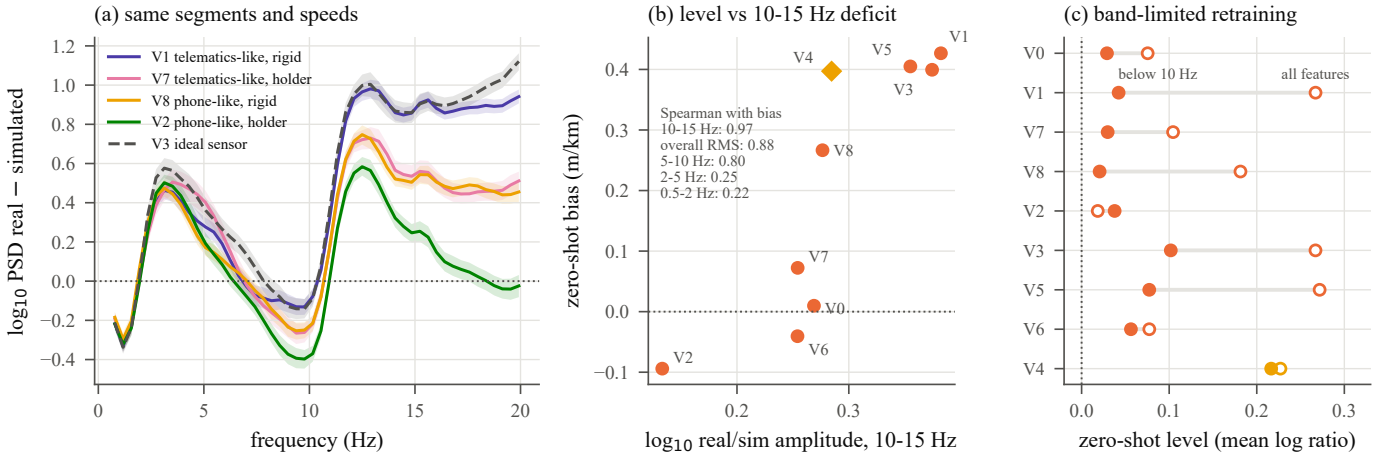


Fig. 2. Simulated versus real vibration. (a) Welch power spectral density, real minus simulated ($\log_{10}$), on the same segments in the same 10 km/h speed bands for four sensor variants and the ideal sensor (bands: 95 % intervals, 1 km-block bootstrap). (b) Zero-shot bias of the nine measured-profile variants against their real/simulated 10–15 Hz amplitude ratio (diamond: V4, one vehicle), with the Spearman correlation of every band. (c) Mean zero-shot log level with all 13 features (open) and with the five features without vibration above 10 Hz (filled).

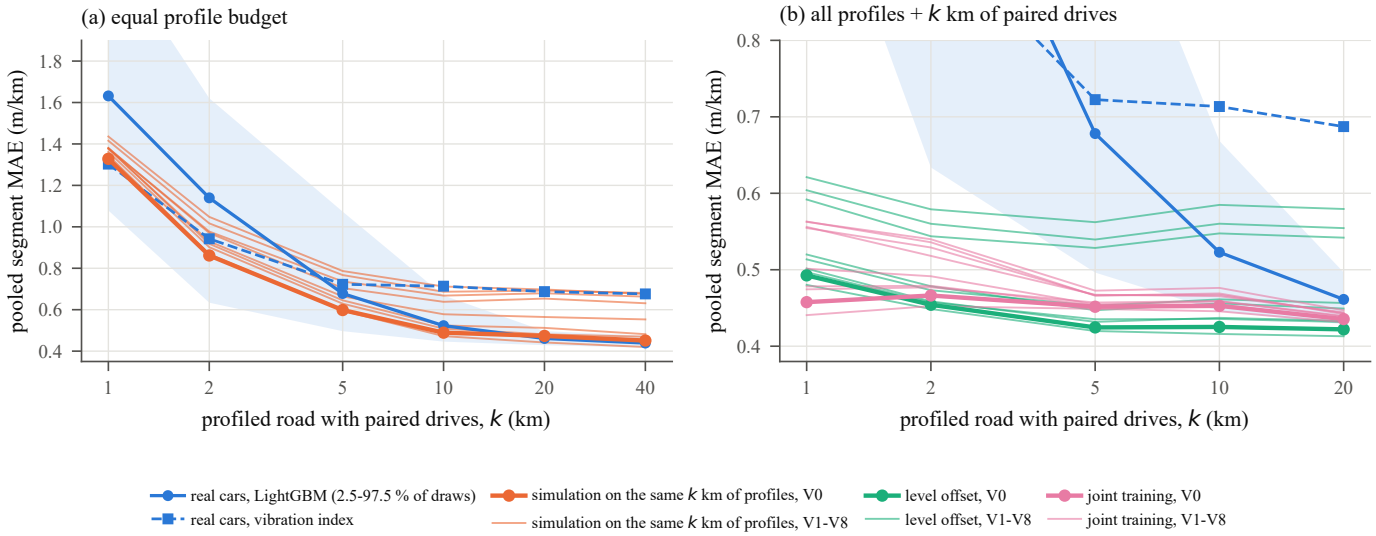


Fig. 3. Simulation against paired real drives on the same 20 draws of k km of training-route road (pooled segment MAE on the held-out routes, means over draws; bold: V0, thin: V1–V8). Blue: real-only LightGBM (band: 2.5–97.5 % of draws) and vibration index (dashed). (a) LightGBM on simulated passes over the profiles of the same k km. (b) Simulation over all training-route profiles combined with the paired passes by a level offset (green) or joint training (pink); real-only values at 1 and 2 km lie above the axis (LightGBM 1.63 and 1.14, index 1.30 and 0.94 m/km).

C. How much paired real data is needed

Fig. 3 compares simulation with real drives on the same draws of k km (every variant and the test-route intervals: Supplement S5). Simulated drives over the profiles of the k km alone are no general substitute for the real drives over them (Fig. 3a): from 5 km on, their value depends on the variant (−0.02 to +0.24 m/km against real-only LightGBM at 20 km). Simulated drives over all profiled training roads, combined with the paired passes, are a substitute (Fig. 3b). With 5 km of paired drives, every variant beats LightGBM trained on those drives alone (pooled MAE 0.68 m/km), with a level offset (0.42–0.56 m/km, better in 13–20 of 20 draws) or with joint training (0.45–0.47 m/km, better in 20 of 20 draws), and the block-bootstrap intervals over the test routes exclude zero for every variant and both methods. The reference simulator with a 5 km offset (0.42 m/km, 0.40–0.45 over draws) is more accurate than 20 km (0.46) or 40 km (0.44 m/km) of real drives alone. One pass per segment suffices: with 49 instead of 475 segment-passes, the simulation-based estimates hardly change (0.43–0.55 and 0.44–0.49 m/km) and stay better than real-only training in every draw, whose error rises to 0.93 m/km. An index with the car's own speed exponent and an affine fit to the same 5 km reaches only 0.78 m/km, so simulation contributes more than a speed normalisation.

With more paired data the two methods diverge. Real-only training overtakes the level offset for V1, V3 and V5, the

variants with the largest 10–15 Hz deficit (the offset wins 0 of 20 draws at 20 km; differences +0.08 to +0.12 m/km), while joint training stays better for every variant (−0.03 to −0.01 m/km at 20 km, 16–20 of 20 draws, all intervals below zero), though not on every route (worse in 5 of 36 variant–route combinations, by at most +0.016 m/km). The reason is the kind of level error: an offset removes a common level and leaves the difference between the motorway and urban levels, which ranges from 0.09 to 0.30 over the variants and orders their MAE after the offset (Spearman 0.88). At 5 km, joint training minus offset is +0.002 to +0.032 m/km for the six variants with small class differences and −0.096 to −0.062 m/km for V1, V3 and V5; with joint training the nine variants span only 0.02 m/km at 5 km and 0.02 m/km at 20 km, against 0.14 and 0.17 m/km with an offset.

IV. The Simulator Fidelity Gap: Game Engines versus Measured Profiles

The source of the road profiles decides whether simulated training transfers (Table I). Driven by measured profiles, the parametric half-car gives the results above. Driven by synthetic ISO 8608 profiles (V9), it still ranks segments ($\rho = 0.81$) but over-predicts motorway roughness by +0.70 m/km. To measure the fidelity gap of game-engine environments, we benchmark a BeamNG.tech [24] corpus: 19 manually driven trips (138 km) of 3 vehicles on 3 maps, logging the vehicle's inertial signals and the ray-cast elevation of the road mesh under both wheel paths, from which IRI labels are computed with the same quarter-car algorithm after the smoothing discussed below. LightGBM trained on it does not transfer to LiRA-CD: its within-route ρ is 0.35 and its MAE 1.26 m/km (0.32 and 2.08 m/km with unsmoothed labels), and its estimates rise steeply with speed on a fixed road (elasticity +0.79).

The ground truth explains much of this. The ray-cast mesh elevation is far rougher than the measured roads, increasingly so at short wavelengths: its displacement power exceeds that of the urban P79 profiles 12-fold at wavelengths of 10–30 m and 42–54-fold at 0.5–5 m (Fig. S2), and its IRI averages 12.5 m/km, against 3.36 m/km on the urban P79 roads. The excess lies in the elevation values, not in their sampling: 54–69 % of the logged readings repeat the previous one while the car moves, but removing the repeats raises the mean IRI to 15.6 m/km. A curvature-adaptive smoothing designed to remove mesh tessellation lowers it to 4.5 m/km, yet applied to the measured profiles it lowers the urban IRI by 28 %, so it cannot repair the labels without distorting real roughness. The logged inertial signals are also degraded: 48–68 % of consecutive samples are exact duplicates. Game engines serve vehicle dynamics and software testing [25], but without measured road profiles they provide neither a valid IRI ground truth nor transferable vibration.

V. Discussion, Recommendations and Limitations

A blueprint for road agencies. For a new car type the results suggest three steps. (i) Simulate the car over corridors whose profiles are already surveyed, with randomised vehicles and a sensor model; the ranking of segments will transfer. (ii) Collect about 5 km of paired drives over profiled road; one pass per segment sufficed here. (iii) Train jointly on simulated and paired drives, which beat real-only training on average for every simulator variant and every k we tested; a level offset is slightly more accurate when the simulated level error is common to both road classes, which the paired drives reveal by class and a comparison of real and simulated spectra on the calibration roads anticipates.

How to evaluate. Because a within-route Spearman correlation is unaffected by calibration, it cannot separate a trained estimator from an untrained statistic; IRI estimators should be judged on held-out routes by absolute error and level per road class. Spatial dependence matters as well: 5 km-block intervals of paired differences were 1.9–2.7 times as wide as segment-level ones, which would have declared the V0-trained model worse than real training. On fixed roads, real-trained estimators over-correct for speed (LightGBM's estimate changes by −13 % between 40 and 100 km/h), presumably because smooth roads are driven fast; simulation-trained LightGBM over-corrects less on urban roads, but this difference is not robust to controlling for within-pass speed variation (Supplement S6).

Limitations. The data cover one car model with one type of telematics unit on four routes in one region, road class and route are confounded, and all intervals describe these routes. Apart from the comparison of the reference simulator with real training, the variant, calibration and spectral analyses were designed after the test routes had been evaluated; they are exploratory and need confirmation on a new car type. The simulator is a linear half-car without roll; its vibration differs from the real car's at 2–5 Hz, above 10 Hz and in its speed scaling. We compared three ways of combining simulated and real data with one learner; network fine-tuning or domain adaptation might rank differently. Transfer to other vehicles was examined only in simulation, where the error on 30 unseen vehicles exceeded that on the training vehicles by +70 % with 3 and +16 % with 30 training vehicles (Supplement S7).

VI. Conclusion

Simulated drives over measured road profiles teach an estimator how a production car's vibration ranks road roughness, but not its absolute level, which depends on the simulated sensor path above 10 Hz and on the simulated vehicle population. About 5 km of paired drives with one pass per segment fix the level: combined with simulation over all profiled roads, they beat real-only training on the same drives for every simulator variant, the reference simulator with a level offset beats 20 km of real drives alone, and joint training keeps an advantage for every variant up to 20 km. Game-engine road meshes cannot replace measured profiles. Whether these requirements hold for other cars and road networks needs new paired data, ideally with the simulator fixed in advance.

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